

UNIT-1-HISTORY OF INDIAN MATHEMATICS

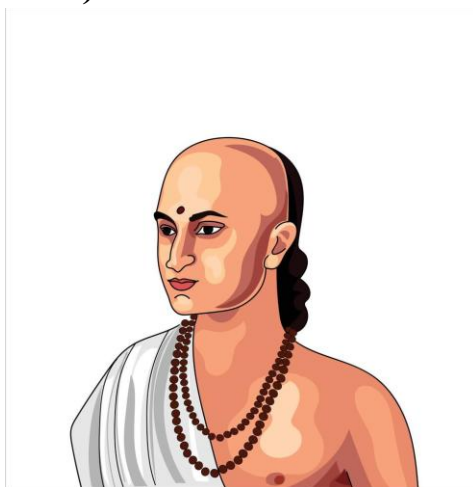
Based on:

1. *Vedic Mathematics* by His Holiness Jagadguru Sankaracharya Swami Bharati Krishna Tirthaji Maharaj (Motilal Banarsidass, 1st Edition, 2022)
2. *Vedic Mathematics Made Easy* by Dhaval Bathia (Jaico Publishing House, 2nd Edition)

1. Famous Mathematicians of India

India has produced some of the most brilliant mathematical minds in the history of the world. These sages(Wise persons or great thinkers.) and scholars made foundational contributions that shaped global mathematical knowledge.

❖ **Aryabhata** (476–550 CE):



- Developed the place-value system and introduced the concept of zero.
- **Aryabhata** was one of the greatest mathematicians and astronomers of ancient India, born in 476 CE, most likely in **Kusumapura (modern-day Patna)**.
- He authored the famous text "**Aryabhatiya**", composed at the age of 23, which covered mathematics and astronomy in a concise poetic style.

- Aryabhatiya is divided into four sections:

1) **Gitikapada**(गीतिकपद) ; where "Gitika" (गीतिका) = a short poem or lyrical verse "Pada" (पद) = line or stanza of a poem/song

2) **Ganitapada**(गणितपद); where "Ganit" (गणित) = Mathematics
"Pada" (पद) = Word / Term

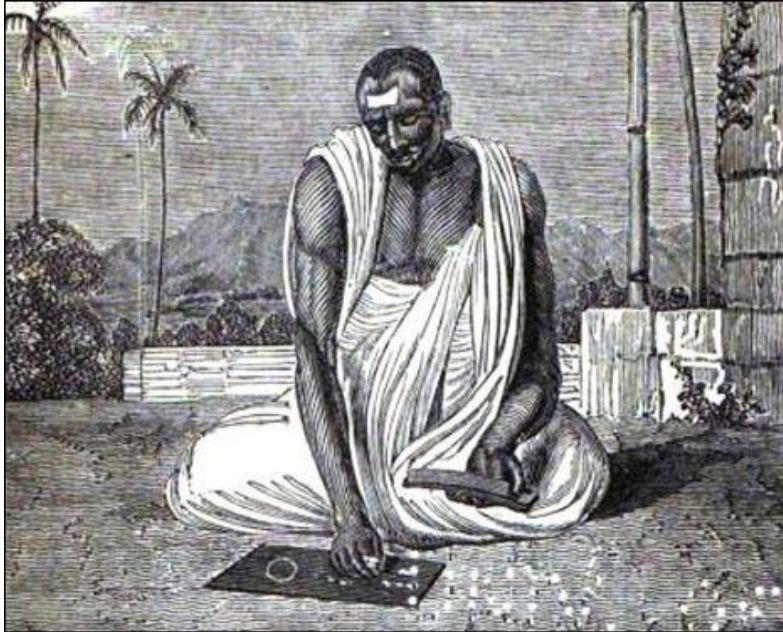
3) **Kalakriyapada**(कालक्रियापद) ; where "Kala" (काल) = Time
"Kriya" (क्रिया) = Action or Verb , "Pada" (पद) = Word

4) **Golapada**(गोलपद) ; where Gola (गोल) = Sphere, Circle, or Globe
, Pada (पद) = Word, Term, Chapter, or Section

- In **Ganitapada**, he introduced **arithmetic, algebra, plane and spherical trigonometry**, and the concept of place value.
- He was the **first to state that Earth rotates on its axis**, explaining the apparent motion of stars.
- Aryabhata **approximated π (pi)** accurately to 3.1416, stating it is irrational — a major advancement for the time.
- He introduced the **concept of zero** and defined rules for **positive and negative numbers**, operations, and square roots.
- Aryabhata proposed a **heliocentric model**, suggesting that the Earth revolves around the Sun, centuries before Copernicus.
(The **heliocentric model** is a scientific theory which states that the **Sun is at the center** of the solar system, and all the **planets revolve around it**.)

- He explained **eclipses scientifically**, rejecting superstitious beliefs; he described lunar and solar eclipses based on shadows.
- He provided accurate formulas to find the **area of a triangle** and the **volume of a sphere**.
- His **sine tables** were among the earliest and used Sanskrit names for trigonometric functions.
- Aryabhata believed in the cyclic nature of time and described **yugas** and the **rotation of heavenly bodies** in detail.
- His mathematical and astronomical works were translated into Arabic, influencing scholars in the Islamic world.
- The **Aryabhata Satellite**, India's first satellite launched in 1975, was named in his honor.
- Aryabhata's legacy continues to inspire generations and reflects the **rich mathematical heritage of ancient India**.

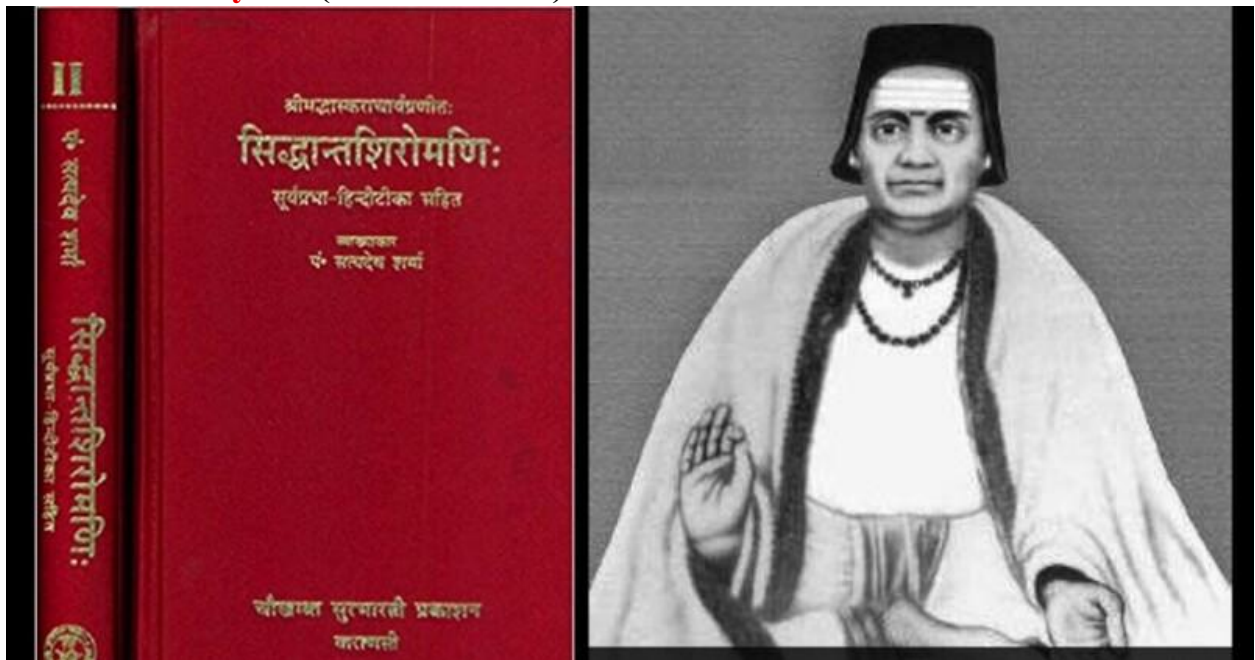
❖ **Brahmagupta (598–668 CE):**



- Introduced rules for zero and negative numbers.
- **Brahmagupta** was a renowned Indian mathematician and astronomer born in 598 CE in **Bhillamala** (modern-day Bhinmal, Rajasthan).
- He authored the famous work "**Brāhmasphuṭasiddhānta**"(ब्राह्मस्फुटसिद्धान्त) in 628 CE at the age of 30.
- This text is one of the earliest to **treat zero as a number**, along with rules for its mathematical operations.
- Brahmagupta defined **rules for addition, subtraction, multiplication, and division** involving both **positive and negative numbers**.
- He introduced concepts of **negative numbers** and **positive numbers** to explain these operations.
- He gave a **formula for the area of a cyclic quadrilateral**, now known as **Brahmagupta's formula**.

- His work includes methods for solving **quadratic equations** and introduced rules for solving **indeterminate equations**.
- Brahmagupta was the first to provide **general solutions to linear and quadratic equations** with negative solutions.
- He gave important rules on arithmetic progressions and **geometric series**.
- As an astronomer, he accurately described **lunar and solar eclipses**, planetary motions, and the length of the year.
- He believed that the Earth was a **sphere** and described **gravity as a force of attraction**.
- Brahmagupta proposed that **time and space are infinite and eternal**, which was revolutionary for his era.
- His influence extended beyond India through Arabic translations that spread to the Islamic world.
- His ideas laid the foundation for **Algebra**, inspiring later mathematicians like Al-Khwarizmi.
- Brahmagupta is regarded as one of the greatest minds in ancient mathematics, with a lasting impact on **both Eastern and Western mathematical traditions**.

❖ **Bhaskaracharya II (1114–1185 CE):**

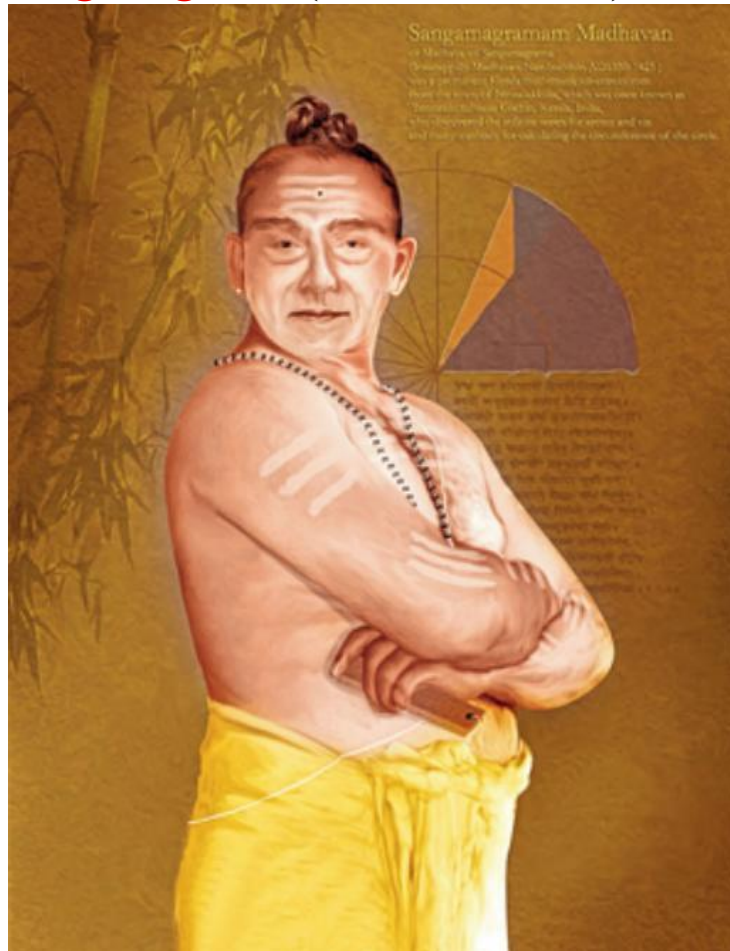


- Authored *Lilavati* (arithmetic) and *Bijaganita* (algebra). His poetic mathematical problems are still admired.
- **Bhaskaracharya II**, also known simply as **Bhaskara II**, was born in **1114 CE** in a village near **Bijapur, Karnataka**.
- He was a brilliant mathematician and astronomer in the tradition of ancient Indian scholars.
- His most famous work is the "**Siddhanta Shiromani**", composed in 1150 CE when he was just 36 years old.
- Siddhanta Shiromani is divided into four parts: **Lilavati (arithmetic)**, **Bijaganita (algebra)**, **Grahaganita (planetary mathematics)**, and **Goladhyaya (sphere or astronomy)**.

- **Lilavati** contains poetic and puzzle-like problems in arithmetic, geometry, and measurement, written for his daughter.
- **Bijaganita** deals with advanced algebra, including solutions to **quadratic, cubic, and indeterminate equations**.
- He was among the first to **recognize that a positive number has two square roots**.
- Bhaskara accurately calculated the **sine of angles** and worked on **trigonometric identities**.
- He introduced concepts similar to **differential calculus**, anticipating parts of Newton and Leibniz's work by centuries.
- He described the concept of **instantaneous motion**, using what is now known as derivatives in modern calculus.
- Bhaskara gave an accurate value of **pi (π)** and worked with **irrational numbers**.
- He described how **gravity** attracts objects downward, echoing thoughts of physical science.

- His astronomical calculations included **planetary positions, eclipses, and conjunctions**.
- Bhaskaracharya's works were translated into Persian and Arabic, influencing global mathematics.
- His legacy lives on as a symbol of **India's advanced mathematical thought and scientific spirit** during the medieval period.

❖ **Madhava of Sangamagrama (c. 1340 – c. 1425):**



- Developed infinite series expansions for trigonometric functions. Precursor of calculus.
- **Madhava of Sangamagrama** was a brilliant Indian mathematician and astronomer born around **1340 CE** in **Sangamagrama** (modern-day Irinjalakuda in Kerala).
- He is regarded as the **founder of the Kerala School of Mathematics and Astronomy**.
- Madhava made groundbreaking contributions to **infinite series, calculus, and trigonometry**, centuries before Newton and Leibniz.

- He was the **first in the world** to develop **infinite series expansions** for trigonometric functions like **sine, cosine, and arctangent**.
- His series for π (pi) is known as the **Madhava–Leibniz series**, expressed as:

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$$
- Madhava calculated the value of π correct to **11 decimal places**, a remarkable achievement for the time.
- He introduced the concept of **convergence** in series and provided correction terms to improve accuracy.
- His work marked the **beginning of calculus in India**, long before it emerged in Europe.
- Madhava accurately determined **sine and cosine values** using infinite series and interpolation.
- He laid the groundwork for later mathematicians like **Nilakantha Somayaji** and **Jyesthadeva**.
- His contributions were compiled in later texts such as **Yuktibhāṣā**("युक्तिभाषा")("तर्कपूर्ण भाषा") and **Tantrasangraha**(तंत्रसंग्रह)("सूत्रों या विधियों का संकलन")
- Madhava also contributed to **astronomical models**, correcting planetary positions and improving almanacs.
- He applied mathematics to **astronomical observations**, such as eclipse predictions and planetary motion.

- His combination of mathematical precision and practical astronomy showed India's advanced understanding of science.
- Madhava is celebrated as a **mathematical genius** who advanced the world's knowledge of **infinite processes and calculus** well before their time.

❖ **Srinivasa Ramanujan (1887–1920):**



- Created deep, intuitive formulas in number theory, continued fractions, and modular functions.
- **Srinivasa Ramanujan** was born on **22 December 1887** in **Erode, Tamil Nadu**, India, into a humble Brahmin family.
- From a young age, he showed an exceptional talent in mathematics, mastering advanced topics without formal training.

- At the age of 16, he obtained a copy of **G.S. Carr's "A Synopsis of Elementary Results"**, which greatly influenced his mathematical development.
- Ramanujan independently discovered complex mathematical theorems in **number theory, infinite series, and continued fractions**.
- Despite limited resources, he created thousands of original results, including **Ramanujan primes, tau function, and modular equations**.
- In 1913, he sent a letter with his work to **G.H. Hardy**, a prominent mathematician at Cambridge University.
- Impressed by Ramanujan's genius, Hardy invited him to **Trinity College, Cambridge**, where Ramanujan worked from 1914 to 1919.
- Ramanujan developed deep results in **partitions, theta functions, elliptic functions, and q-series**.
- He formulated over **3,900 original identities and equations**, many of which were later proven true.
- One of his famous contributions is the **Ramanujan-Hardy number 1729**, the smallest number expressible as the sum of two cubes in two ways.
- Ramanujan was deeply spiritual and credited his mathematical intuition to **goddess Namagiri** of Namakkal.
- He was elected as a **Fellow of the Royal Society (FRS)** in 1918, a rare honor for a self-taught Indian mathematician.
- Due to poor health and unfamiliar climate, he returned to India in 1919 and passed away at the young age of **32 in 1920**.

- His last writings, known as the “**Lost Notebook**”, contained many theories that are still being explored and applied in modern mathematics and physics.
- Ramanujan is remembered as a **mathematical genius**, whose intuition and creativity continue to inspire mathematicians around the world.

❖ **Swami Bharati Krishna Tirthaji Maharaj (1884–1960):**



- Rediscovered Vedic Mathematics from ancient texts, systematized into 16 Sutras and 13 Upasutras.
- **Swami Bharati Krishna Tirthaji Maharaj** was born in **March 1884** in Tamil Nadu, India, as **Venkatraman**.
- He was a brilliant student who mastered **Sanskrit, Mathematics, History, Philosophy, and English** at a young age.

- He completed his education at **Madras Christian College** and later became a respected scholar and teacher.
- He was deeply influenced by Indian scriptures and spiritual traditions, leading him to adopt the monastic life.
- After becoming a **sanyasi**, he was ordained as the **Shankaracharya of Govardhan Matha, Puri**, one of the highest religious positions in Hinduism.
- Between **1911 and 1918**, through intense study and meditation on the **Atharvaveda**, he rediscovered **Vedic Mathematics**.
- He compiled the system into **16 Sutras (formulas)** and **13 Upasutras (sub-formulas)** that simplify complex mathematical operations.
- These sutras cover topics like **arithmetic, algebra, geometry, trigonometry**, and even **calculus**.
- He believed that Vedic Mathematics was not his invention, but a **rediscovery of ancient wisdom** encoded in the Vedas.
- Swamiji delivered lectures across India and abroad, including in **the USA and UK**, to promote Vedic Mathematics.
- His teachings emphasized **mental math, clarity of thinking, and simplicity in calculations**, making math joyful and intuitive.
- He worked tirelessly to revive India's ancient mathematical heritage and connect it with spiritual philosophy.

- His book “**Vedic Mathematics**” was published posthumously in **1965** by Motilal Banarsidass.
- Swamiji passed away in **1960**, but his legacy continues through scholars, teachers, and math enthusiasts globally.
- Today, he is revered as the **pioneer who bridged ancient Indian wisdom with modern mathematical education.**

2. Mathematics in the Vedas

The four Vedas—**Rigveda, Yajurveda, Samaveda, and Atharvaveda**—contain hymns (religious songs or poems), rituals, and philosophical discussions. Hidden within these are subtle (not obvious, or difficult to understand.) mathematical principles, expressed through language, symbolism, and numerical ideas.

- The **Śulbasūtras**, part of the appendices to the Vedas, contain geometric principles used in altar construction, such as the Pythagorean theorem.
- Concepts of **permutations, combinations, and fractions** were used in organizing chants and rituals.
- Large cyclic time measurements (Yugas, Kalpas) show an understanding of astronomical mathematics.

Swamiji interpreted Vedic sutras as **mathematical formulae**—brief, intuitive, and powerful tools for performing calculations.

3. Introduction to Vedic Mathematics

Vedic Mathematics is a mental system of mathematics based on 16 main sutras and 13 upa-sutras . These principles cover topics such as:

- Arithmetic (addition, subtraction, multiplication, division)
- Algebra (factorization, equations, squaring)

- Geometry and trigonometry
- Calculus (basic integration/differentiation forms)

The core philosophy is **simplicity and mental agility**. This system enables students to:

- Calculate quickly
- Perform mental math
- Understand numbers better
- Reduce dependency on calculators

❖ **Indian Kalganana (Time Calculation)**

Indian mathematicians made significant contributions to calendar science and time computation:

- **Tithi (lunar day), Paksha (fortnight), Masa (month), and Samvatsara (year)** are based on lunar and solar cycles.
- Tools like the **Ghatika Yantra** were used for time measurement.



- **Yugas and Kalpas** represented massive time spans (1 Mahayuga = 4,320,000 years).

- These calculations required knowledge of fractions, sequences, and large-number arithmetic.

Swamiji illustrates how Vedic techniques simplify such large-scale computations through **Nikhilam** and **Anurupyena** methods.

4. Main Sutras of Vedic Mathematics (16 Sutras)

These are **short** statements from Sanskrit, each offering a shortcut to mathematical operations. Below are selected sutras:

1. **Ekadhikena Purvena (एकाधिकेन पूर्वेण)** – By one more than the previous one

General Formula:

For any number ending in 5, say:

Number = $x5$

Let the part before 5 be x . Then:

$(x5)^2 = x \times (x + 1)$ and write 25 at the end.

Example 1: Find 25^2 .

Step-by-step:

- Number before 5 is **2**
- Increase it by 1: **$2 + 1 = 3$**
- Multiply: **$2 \times 3 = 6$**
- Place **25** at the end

Answer: $25^2 = 625$

Example 2: Find 75^2

- Number before 5 is **7**

- Add 1: $7 + 1 = 8$
- Multiply: $7 \times 8 = 56$
- Append 25

Answer: $75^2 = 5625$

Example 3: Find 105^2

- Number before 5 is **10**
- Add 1: $10 + 1 = 11$
- Multiply: $10 \times 11 = 110$
- Append 25

Answer: $105^2 = 11025$

➤ Benefits of Using This Sutra:

- Mental calculation without paper
- Fast and easy squaring of numbers ending in 5
- Useful in competitive exams

2. **Nikhilam Navatashcaramam Dashatah** (निखिलम् नवतश्चरमं दशतः) – All from 9 and the last from 10

This sutra is one of the most powerful and widely used in **Vedic Mathematics**, particularly for:

- **Subtraction** from powers of 10

Use in Subtraction from Base:

Example: Find $1000 - 783$ using this sutra

- Apply: All from 9 and last from 10
- $7 \rightarrow 9 - 7 = 2$
- $8 \rightarrow 9 - 8 = 1$

- $3 \rightarrow 10 - 3 = 7$

Answer: $1000 - 783 = 217$

3. **Urdhva-Tiryagbhyam (उर्ध्व-तिर्यग्भ्याम्)** – Vertically and crosswise

This is one of the most important and general sutras in Vedic Mathematics. It is used for multiplication of all types — small or large numbers, even algebraic expressions.

Let's begin with a 2-digit $\times$ 2-digit example:

Example: 12×13

Step-by-step (Write numbers like this):

$$\begin{array}{r} 12 \\ \times 13 \\ \hline \end{array}$$

Step 1: Multiply the units digits

$$2 \times 3 = 6 \rightarrow \text{Rightmost digit}$$

Step 2: Cross-multiply and add:

$$(1 \times 3) + (2 \times 1) = 3 + 2 = 5$$

Step 3: Multiply left digits (tens)

$$1 \times 1 = 1$$

Now write the digits in order: 1 5 6

Answer: $12 \times 13 = 156$

Example: 23×34

$$\begin{array}{r} 23 \\ \times 34 \\ \hline \end{array}$$

Step 1: $3 \times 4 = 12 \rightarrow$ write 2, carry 1

Step 2: $(2 \times 4) + (3 \times 3) = 8 + 9 = 17 + 1(\text{carry}) = 18 \rightarrow$ write 8, carry 1

Step 3: $2 \times 3 = 6 + 1(\text{carry}) = 7$

Answer: 782

4. **Paravartya Yojayet** (परावर्त्य योजयेत्) – Transpose and adjust

The Vedic Mathematics Sutra **Parāvartya Yojayet** (परावर्त्य योजयेत्) means "**Transpose and Adjust**". It is mainly used for division, especially when the divisor is slightly more or less than a base like 10, 100, 1000, etc.

Example 1: Base 10

Divide 123 by 9

Divisor: 9 $\rightarrow$ Base 10, Deficit -1

Parāvartya: +1

Work from left of dividend:

123

↓

Step 1: bring down 1 $\rightarrow$ 1

Step 2: $1 \times 1 = 1 \rightarrow$ add to next digit: $2 + 1 = 3$

Step 3: $3 \times 1 = 3 \rightarrow$ add to next digit: $3 + 3 = 6$

Quotient: 13, Remainder: 6

Example 2: Base 100

Divide 1005 by 98

Divisor: 98 $\rightarrow$ Base 100, Deficit -2 $\rightarrow$ +2

Dividend: 1005 $\rightarrow$ split as 10 | 05

Step 1: Bring down 10

Step 2: $10 \times 2 = 20 \rightarrow$ add to 05: $05 + 20 = 25$

Quotient: 10, Remainder: 25

Example 3: Base 100

Divide 1234 by 101

Divisor: 101 $\rightarrow$ Base 100, Surplus +1 $\rightarrow$ write -1

Dividend: 1234 $\rightarrow$ 12 | 34

Step 1: Bring down 12

Step 2: $12 \times (-1) = -12 \rightarrow$ add to 34: $34 - 12 = 22$

Quotient = 12, Remainder = 22

Example 4: Base 1000

Divide 10405 by 999

Dividend: 10 | 405 (Base 1000)

Divisor: 999 $\rightarrow$ Base 1000, Deficit -1 $\rightarrow$ +1

Step 1: Bring down 10

Step 2: $10 \times 1 = 10 \rightarrow$ add to 405+10= 415

Quotient = 10, Remainder = 415

When to Use Parāvartya Yojayet:

- Divisors slightly more or less than base values (like 10, 100, 1000)
- Long divisions with recurring decimals

5. **Shunyam Samyasamuccaye (शून्यं सम्यसमुच्चये)** –

"If the Samuccaya is the same, it is zero."

;Where *Samuccaya* means **collection, total, or aggregate**.

Example 1: Solve $\frac{x+2}{x+3} = \frac{y+2}{y+3}$

- Samuccaya (numerator + denominator):
Left: $x + 2 + x + 3 = 2x + 5$
Right: $y + 2 + y + 3 = 2y + 5$
Coefficients (2 and 5) same on both sides.
- Since **Samuccaya is same**, apply: $x = y$

Answer: $x = y$

Example 2: Solve $\frac{a^2+ab}{a+b} = \frac{c^2+cb}{c+b}$

- Samuccaya:
Numerator = $a^2 + ab$,
Denominator = $a + b$
Right side has same form with c instead of a .
Samuccaya is identical
So by sutra $a = c$

Example 3: Solve $\frac{3x+2y+7}{5x+y+1} = \frac{3a+2b+7}{5a+b+1}$

- Check Samuccaya:
LHS total terms: $3x + 2y + 5x + y + 7 + 1 = 8x + 3y + 8$
RHS total terms: $3a + 2b + 5a + b + 7 + 1 = 8a + 3b + 8$
Both sides: **Same structure**

Apply sutra directly

$$x = a, y = b$$

6. **Anurupye Shunyamanyat (अनुरूप्ये शून्यमन्यत्) –**

Meaning:

"If one is in proportion, the other is zero."

Understanding the Sutra:

This sutra is used mainly in solving **simultaneous linear equations** (especially two-variable ones) where the **coefficients are in proportion**.

Application Condition:

When two equations are in such a way that the **ratios of the coefficients of two variables are equal**, but the ratio of the constants is **not equal**, then **one variable must be zero**.

Mathematical Form:

Given two equations:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

If:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Then by **Anurupye Shunyamanyat**:

One of the variables (x or y) is **zero**.

Example 1 :Solve:

$$2x + 3y = 6$$

$$4x + 6y = 15$$

Step-by-Step:

- Check ratio of coefficients:

$$\frac{2}{4} = \frac{3}{6} = \frac{1}{2}$$

- Check ratio of constants:

$$\frac{6}{15} = \frac{2}{5} \neq 12$$

Coefficient ratios are same

Constant ratio is different

Apply Sutra $\Rightarrow$ One variable is zero

case(1) :Let's try $x = 0$:

$$2(0) + 3y = 6 \Rightarrow y = 2$$

Satisfies first equation. Check second:

$$4(0) + 6(2) = 12 \neq 15$$

Doesn't work.

Case(2): Let's try $y = 0$:

$$2x + 0 = 6 \Rightarrow x = 3$$

Check second:

$$4(3) + 0 = 12 \neq 15$$

Still not working.

Conclusion: When the ratio of coefficients is equal, and constants are not, **no consistent solution exists** unless one variable is zero and satisfies both.

So, there is no solution for this given system.

Example: Solve:

$$3x + 6y = 12$$

$$6x + 12y = 18$$

Step-by-Step:

- Check ratio of coefficients:

$$\frac{3}{6} = \frac{6}{12} = \frac{1}{2}$$

- Check ratio of constants:

$$\frac{12}{18} = \frac{2}{3} \neq \frac{1}{2}$$

Coefficient ratios are same

Constant ratio is different

Constant ratio is different

Apply Sutra $\Rightarrow$ One variable is zero

case(1) :Let's try $x = 0$:

$$3(0) + 6y = 12$$

So, $y = 2$

Put $y = 2$ in $6x + 12y = 6(0) + 12(2) = 24 \neq 18$

Doesn't work.

Case(2): Let's try $y = 0$:

$$3x + 6(0) = 12$$

So, $x = 4$

Put $x = 4$ in $6x + 12y = 6(4) + 12(0) = 24 \neq 18$

So, given system of equation has no solution.

7. **Sankalana-vyavakalanabhyam (संकलन-व्यवकलनाभ्याम्)** – By addition and by subtraction

Add the same number to both minuend and subtrahend to simplify subtraction. (Minuend – Subtrahend = Difference)

Steps:

- Add a number to both minuend and subtrahend to make the subtraction easier
- Commonly used to remove zeros or simplify borrow

Examples:

1. $5002 - 1897$

- Add 3 to both: $(5005 - 1900) = 3105$
- 2. $4000 - 1998$
 - Add 2: $(4002 - 2000) = 2002$
- 3. $7001 - 3996$
 - Add 4: $(7005 - 4000) = 3005$
- 8. **Puranapurāṇabhyāsa** (पूरणापूरणाभ्याम्)– By the completion or non-completion

The Sutra **Puranapurāṇabhyāsa** is a versatile tool in Vedic Mathematics. It is used in cases where **completion (Pūrāṇa)** or **non-completion (Apūrāṇa)** of numbers is involved—especially in arithmetic operations, algebraic manipulations, or squaring.

This technique is often applied in the following situations:

his sutra is used in:

- **Multiplication** of numbers close to a base (either more or less)
- **Completing numbers to a base for faster calculation**

Example 1: Using Complements for Multiplication

Multiply 98×97 using Puranapurāṇabhyāsa.

Step 1: Base = 100

98 is 2 less than 100 $\rightarrow -2$

97 is 3 less than 100 $\rightarrow -3$

Step 2:

- Left part: $98 - 3 = 95$ (or $97 - 2$)
 - Right part: $(-2) \times (-3) = 6$
- So answer = **9506**

Answer: 9506

5. Upasutras (13 Corollaries)

These are supportive rules that enhance or simplify the application of the main sutras:

1. **Anurupyena (अनुरूप्येण)** – Proportionately

The sutra "**Anurupyena**" is generally used when the number is **not close to a base**, but to a **proportional base**, i.e., a **multiple or sub-multiple** of a convenient base like 10, 100, 1000 etc.

It helps in simplifying calculations like:

- Multiplication of numbers near **non-standard bases**
- Squaring numbers that are not near powers of 10
- Applying **proportional adjustment** to intermediate results

This sutra allows **changing the base** for easier computation and then **adjusting the answer proportionately**.

Example 1: Multiplication Near a Proportional Base

Multiply:

$$48 \times 42$$

These are not near 10, 100, or 1000.

But 50 (a base multiple of 10) is a better fit. Let's take base = **50**

Step 1:

Let base = 50 (which is **half of 100** → so **adjustment factor = 2**)

- $48 = 50 - 2$
- $42 = 50 - 8$

Cross-subtract:

- $48 - 8 = 40$

- Right part: $(-2) \times (-8) = 16$

Now we have:

Left part = 40, Right part = 16

So answer (using base 50):

$$= 40 \mid 16$$

$$= 40 \times 50 + 16 = 2000 + 16 = \mathbf{2016}$$

Answer: 2016

Example 2: Squaring a Number Near a Multiple Base

Square **53** using base = **50**.

- $53 = 50 + 3$
- Step 1: Left part = $53 + 3 = 56$
- Step 2: Right part = $3^2 = 9$

Answer on base 50:

$$= 56 \mid 09$$

$$= 56 \times 50 + 9 = 2800 + 9 = \mathbf{2809}$$

Answer: 2809

Example 3: Multiplication Using a Sub-base

Multiply **104** $\times$ **108**

These are near base **100**, so we don't need "Anurupyena" here. But suppose we were multiplying **1040** $\times$ **1080**, we might consider base **1000**, or proportional base **1100**, and apply **Anurupyena** to scale the result accordingly.

2. **Yavadunam Tavadunikritya Varga Yojayet (यावदूनं तावदूनिकृत्य वर्गं योजयेत्)**

– "Whatever the deficiency, subtract that from the number, and add the square of the deficiency."

Example 1: Square of 96 (near base 100)

- Base = 100
- Deficiency = $100 - 96 = 4$

Step 1:

$$\text{Left part} = 96 - 4 = 92$$

Step 2:

$$\text{Right part} = 4^2 = 16$$

$$\text{Final Answer} = \mathbf{9216}$$

Example 2: Square of 98

- Base = 100
- Deficiency = $100 - 98 = 2$

$$\text{Left part} = 98 - 2 = 96$$

$$\text{Right part} = 2^2 = 4 = 04$$

$$\text{Final Answer} = \mathbf{9604}$$

Example 3: Square of 994 (base 1000)

- Deficiency = $1000 - 994 = 6$

$$\text{Left part} = 994 - 6 = 988$$

$$\text{Right part} = 6^2 = 36$$

But since base is 1000, we need **3 digits** on the right.

$$\text{So right part} = \mathbf{036}$$

$$\text{Final Answer} = \mathbf{988036}$$

3. **Sopaantyadvayamantyam(सोपान्त्यद्वयमन्त्यम्)** – The last two terms together

(“Sopa-antya-dvaya-mantya” breaks into: *Saha* = with, *antya* = last, *dvayam* = two, *antyam* = last)

The **Sutra "Sopaantyadvayamantyam"** is used in **special multiplication cases**, particularly when numbers follow certain patterns.

This method focuses on using:

- The **last digit** (antyam) sum is 10.
- And **twice the second-last digit** (penultimate)
- Often used in **algebra** and **special number multiplication**

Example 1: Multiplication of Numbers with Same Tens Digit and Units Digit Adding to 10

Let's multiply 43×47

- Both have same tens digit: 4
- Units digits: $3 + 7 = 10$

Apply the sutra:

Step 1:

Multiply tens digit by next number:

$$4 \times 5 = 20$$

Step 2:

Multiply units digits:

$$3 \times 7 = 21$$

So, answer = **2021**

$$43 \times 47 = 2021$$

This is a perfect example of **Sopaantyadvayamantyam**, where the focus is on:

- Last two digits multiplication
- Use of second last digits in a fixed pattern

4. **Ekanyunena Purvena** (एकान्यूननेन पूर्वेण) – One less than the previous

This sutra is primarily used for **multiplication of numbers consisting of all 9s**, such as:

- 9, 99, 999, 9999, etc.
and a **smaller number immediately below** these bases, like:
- 8×9
- 97×99
- 999×997

The sutra tells us to:

1. **Take one less than the left-hand number** (i.e., Ekānyūna – “one less”)
2. **Multiply the result by the base number’s leading digit(s)**
3. **Subtract the right-hand number from 100, 1000, etc.** (as appropriate) and multiply

Example 1: 9×8

Use base = 10 (since 9 is one less than 10)

- One less than 8 = 7
- Subtract 8 from 10 = 2
- Answer = 7 | 2 $\rightarrow$ 72

Answer: $8 \times 9 = 72$

Example 2: 99×97

Use base = 100

- One less than 97 = 96
- Subtract 97 from 100 = 3
- Multiply: 96 | 03 $\rightarrow$ 9603

Answer: $97 \times 99 = 9603$

Example 3: 999×994

Use base = 1000

- One less than 994 = **993**
- Subtract 994 from 1000 = **6**
- Square or multiply: $993 \mid 006 \rightarrow \mathbf{993006}$

Answer: $994 \times 999 = 993006$

Example 4: 9999×9996

Use base = 10000

- One less than 9996 = **9995**
- Subtract 9996 from 10000 = **4**
- Right part = $4^2 = 16 \rightarrow$ Needs 4 digits $\rightarrow 0004 \times 4 = \mathbf{0004}$
- Answer = $9995 \mid 0004 \rightarrow \mathbf{99950004}$

Answer: $9996 \times 9999 = 99950004$

5. Usefulness and Modern Applications

Vedic Mathematics offers immense benefits in various fields:

- **Education:** Enhances speed and accuracy, builds number sense
- **Competitive Exams:** Reduces time for calculations in exams like CAT, GRE, SSC, etc.
- **Daily Life:** Makes budgeting, shopping, and banking easier

❖ ASSIGNMENT/ QUESTION BANK

Section A: Short Answer Questions (2–3 marks each)

1. Explain the contribution of Aryabhata to Indian mathematics.
2. What are the four parts of Aryabhatiya? Briefly describe each.
3. What is the importance of Brahmagupta's work *Brāhmasphuṭasiddhānta*?
4. How did Bhaskaracharya's work in *Lilavati* and *Bijaganita* influence mathematics?
5. Who was Madhava of Sangamagrama and what was his contribution to calculus?
6. Explain the mathematical achievements of Srinivasa Ramanujan in number theory.
7. Discuss the role of Swami Bharati Krishna Tirthaji Maharaj in rediscovering Vedic Mathematics.
8. What is the significance of sine tables in Aryabhata's work?
9. How did Brahmagupta handle negative numbers and zero?
10. What were the key features of the Kerala School of Mathematics?

Section B: Vedic Sutra-Based Questions (3–5 marks each)

11. Explain the sutra *Ekadhikena Purvena* with two step-by-step examples.
12. Describe the use of *Nikhilam Navatashcaramam Dashatah* in subtraction with an example.
13. State the meaning and application of the sutra *Urdhva Tiryagbhyam* in multiplication. Give an example.
14. How does the sutra *Paravartya Yojayet* simplify division? Illustrate with a base 100 example.
15. Explain the application of *Shunyam Samyasamuccaye* in solving algebraic expressions. Provide one example.
16. What is the condition for applying the sutra *Anurupye Shunyamanyat* in linear equations? Give a suitable example.
17. How does *Sankalana-Vyavakalanabhyam* make subtraction easier? Give real-life arithmetic examples.
18. Illustrate the sutra *Yavadunam Tavdunikritya Varga Yojayet* with two examples of squaring numbers near a base.

19. What is the use of the sutra *Sopaantyadvayamantyam* in special multiplication patterns? Explain with an example.
20. Describe how the sutra *Ekanyunena Purvena* is used for multiplying numbers like 99×97 .

Section C: Conceptual & Analytical Questions (5 marks each)

21. Discuss the mathematical significance of the Vedas and Sulbasutras in ancient Indian culture.
22. What is meant by the “place-value system”? How did Aryabhata and Indian mathematicians contribute to its development?
23. Describe how Vedic Mathematics helps in fast mental calculations with modern-day examples.
24. Compare the ancient Indian concepts of time (Yuga, Kalpa) with modern time calculation.
25. Explain the modern-day applications of Vedic Mathematics in education and competitive exams.